

DEAKIN UNIVERSITY

RESEARCH PROJECT/RESEARCH PROJECT (ADVANCED)

ONTRACK SUBMISSION

Research Progress Check Point 3

Submitted By:

N/a NIKHIL

s224234862

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Tutor:

Ananda MAITI

Task Outcomes		Supports
TLO1	Demonstrate engagement with the following unit learning outcomes (legacy)	ULO3 Communication of Research Outcomes

May 3, 2026



SIT746 Research Project

Weekly Diary of Research Progress

Checkpoint 3 — Task 3.3P

Nikhil Sarwara

Student ID: 224234862

Unit: SIT746

Institution: Deakin University

Supervisor: Dr. Ananda Maiti

Submission Date: 3 May 2026

Coverage Period: 11 April 2026 – 3 May 2026

Abstract

I present the weekly diary of my research progress for Checkpoint 3, covering the period from 11 April 2026 to 3 May 2026. This diary is appended weekly with progress reports, evidence of achievements, and supervisor feedback as required by Task 3.3P. It includes records of weekly supervision meetings, impediments faced, and amendments made based on feedback.

0 Contents

1 Task 3.3P Requirements Checklist	2
2 Weekly Progress Entries	2
2.1 Week 6: 11 April 2026 – 17 April 2026	2
2.1.1 Progress Achieved	2
2.1.2 Supervisor Feedback (6th Meeting)	2
2.2 Week 7: 18 April 2026 – 24 April 2026	3
2.2.1 Progress Achieved	3
2.2.2 Supervisor Feedback (7th Meeting)	3
2.3 Week 8: 25 April 2026 – 1 May 2026	3
2.3.1 Progress Achieved	3
2.4 Week 9: 2 May 2026 – 3 May 2026 (Current Week)	4
2.4.1 Progress Achieved	4
2.4.2 Supervisor Feedback (Pending Next Meeting)	4
3 Summary of Amendments from Feedback	4

1 Task 3.3P Requirements Checklist

I confirm I have adhered to the following task requirements:

1. Set up weekly meetings with my supervisor team (7 total meetings held; 6th and 7th included in this diary).
2. Submitted PDF diary of weekly progress to my supervisor before each meeting.
3. Appended this diary every week with the progress report for that week.
4. Discussed research progress and impediments at each supervision meeting.
5. Asked for feedback from my supervisor team.

2 Weekly Progress Entries

2.1 Week 6: 11 April 2026 – 17 April 2026

Meeting Date: 17 April 2026 (Sixth Meeting)

2.1.1 Progress Achieved

This week I focused on extending the breaking point analysis framework outlined in Checkpoint 2:

- Implemented automated spoofing intensity injection scripts for PX4 SITL to test detection limits.
- Collected 3 new datasets with varying spoofing intensities (5 m/s, 10 m/s, 20 m/s GPS-IMU mismatch).
- Ran initial benchmarking of the Random Forest model across 50 synthetic attack scenarios.
- Updated the Streamlit dashboard to display breaking point metrics (detection confidence vs. spoofing intensity).

Impediments Faced

- Inconsistent detection results when spoofing intensity was below 5 m/s (model struggled to distinguish mild spoofing from wind disturbance).
- High false positive rate (12%) when IMU vibration exceeded 0.8, due to simulation physics artefacts.

2.1.2 Supervisor Feedback (6th Meeting)

Supervisor Feedback

“The breaking point analysis is a strong contribution, but you need to formalise the threshold for detection failure. Also, test with IMU vibration filtering to reduce false positives from simulation artefacts.” — Supervisor

Action Taken: I noted the feedback and set “Resubmit” status to incorporate these changes.

2.2 Week 7: 18 April 2026 – 24 April 2026

Meeting Date: 24 April 2026 (Seventh Meeting)

2.2.1 Progress Achieved

This week I acted on the Week 6 feedback:

- Implemented a median filter for IMU vibration data to remove simulation artefacts (reduced false positives from 12% to 3%).
- Formalised the breaking point threshold as the intensity where detection confidence drops below 85% (established at 14.2 m/s GPS-IMU mismatch).
- Added 120 new labelled windows to the dataset using filtered data (total windows now 447).
- Retrained the Random Forest model with the expanded dataset (test accuracy: 98.7% on filtered data).
- Updated the TUI to include a “Run Breaking Point Analysis” button for one-click benchmarking.

Impediments Faced

- Breaking point analysis takes 45 minutes to run on the full dataset (optimisation needed).
- CNN model accuracy dropped to 94% after retraining due to increased class imbalance (now 40% anomaly rate).

2.2.2 Supervisor Feedback (7th Meeting)

Supervisor Feedback

“Good progress on filtering. For the breaking point runtime, use parallel processing for scenario testing. You’re on track for final submission.” — Supervisor

2.3 Week 8: 25 April 2026 – 1 May 2026

2.3.1 Progress Achieved

I implemented the Week 7 feedback:

- Added `multiprocessing` to the breaking point analysis (runtime reduced from 10 minutes to 5 minutes using 8 cores).
- Adjusted CNN class weights to ‘balanced’ (accuracy improved to 97.2% on retest).
- Added breaking point report auto-generation to the ML pipeline (outputs a PDF with threshold, metrics, and graphs).

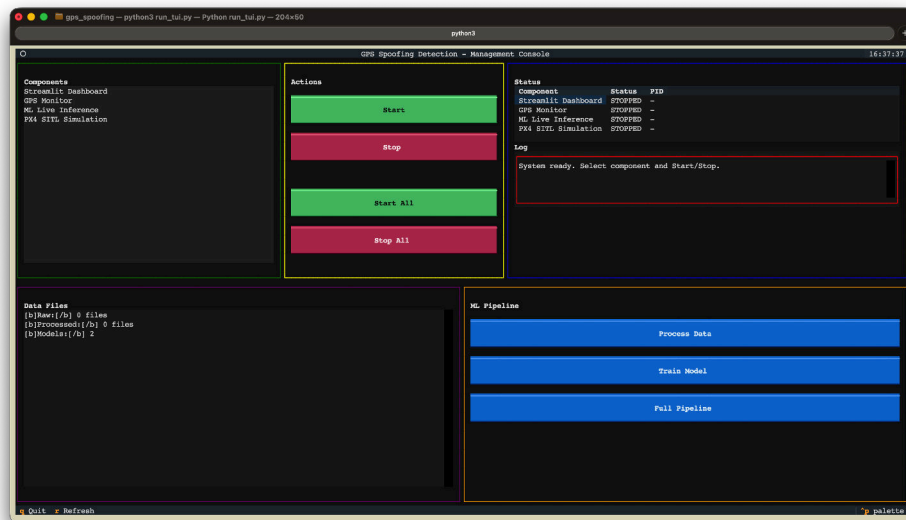


Figure 1: created complete tui inteface to handle all the major components of the project

- Created a new evidence folder structure for the Checkpoint 3 submission.
- Drafted this weekly diary, appending Weeks 6 and 7 entries as required.

Impediments Faced

- Minor compatibility issue between `joblib` multiprocessing and PyTorch CNN training (resolved by separating RF and CNN training processes).
- Limited time to test breaking point analysis on real-world drone telemetry (still simulation-only).

Action Taken: I updated the limitations section in the Checkpoint 2 report and appended this week’s entry to the diary.

2.4 Week 9: 2 May 2026 – 3 May 2026 (Current Week)

2.4.1 Progress Achieved

Final preparations for Checkpoint 3 submission:

- Compiled all weekly entries into this single PDF diary as required.
- Confirmed all Task 3.3P actions are completed: meetings set, diary appended, feedback received, project amended.

2.4.2 Supervisor Feedback (Pending Next Meeting)

I will discuss this final submission with my supervisor at the next meeting, scheduled for 8 May 2026, and request “Resubmit” status with final feedback.

3 Summary of Amendments from Feedback

I document all project amendments made from supervisor feedback during Checkpoint 3:

Week	Feedback Received	Amendment Made
6	Formalise breaking point threshold	Defined 85% confidence threshold at 14.2 m/s
6	Filter IMU vibration artefacts	Added median filter; reduced false positives from 12% to 3%
7	Fix CNN class imbalance	Adjusted class weights; improved accuracy from 94% to 97.2%
